

WISH - 2026 - Voltage Sources and RC Circuits

* Full Name

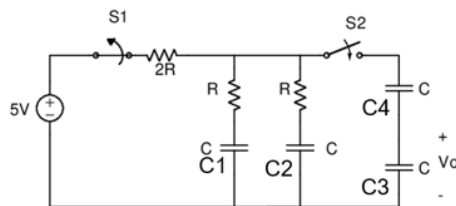
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WiSH26007

Q1. Switch S1 is opened at $t=0$ and S2 is closed.
What is the value of V_o at steady state.
 $R = 100 \text{ ohms}$, $C = 1 \text{ F}$.

- a) 2V
- b) 2.5V
- c) 4V
- d) 5V



Answer the Question 1

☐ **a**

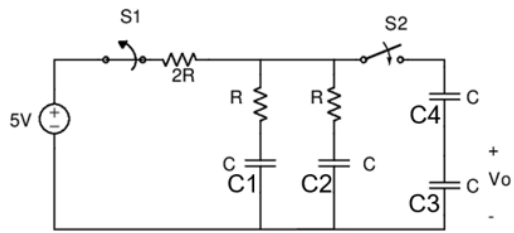
☐ c

☐ b

☐ d

Q2. What is the total energy dissipated in each of the resistor valued R
Switch S_1 is opened at $t=0$ and S_2 is closed. $R = 100 \text{ ohms}$, $C = 1\text{F}$.

- a) 0.5J
- b) 1J
- c) 2J
- d) 4J



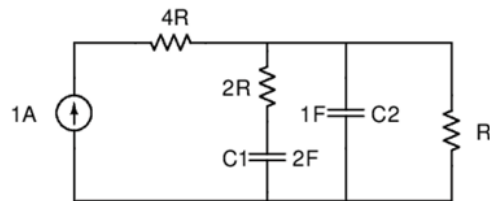
Answer the Question 2

- ☒ **a**

☐ c
- ☐ b
 ☐ d

Q3. What is the charge accumulated on capacitor C_1 in steady state. Value of R is 10 ohms .
 $C_1 = 2\text{F}$ and $C_2 = 1\text{F}$.

- a) 10 Coulombs
- b) 20 Coulombs
- c) 40 Coulombs
- d) 5 Coulombs



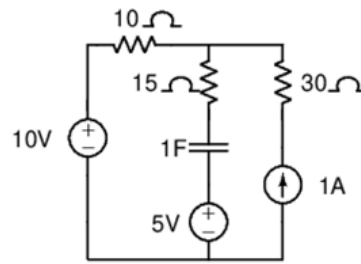
Answer the Question 3

- ☐ a
 ☐ c
- ☒ **b**

☐ d

Q4. After the circuit has reached its steady state, what is the power dissipated the 1A current source in the following figure.

- a) 30 W
- b) 20 W
- c) 40 W
- d) 50 W



Answer the Question 4

☐ a

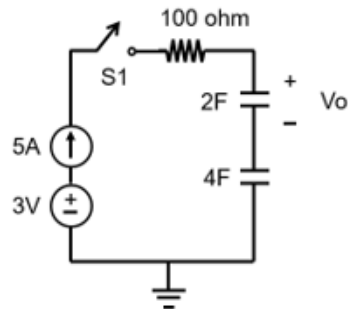
☐ c

☐ b

☒ d

Q5. Switch S1 is closed at $t=0$. What is the value of V_o at $t=1s$.

- a) 2V
- b) 2.5V
- c) 3V
- d) 5V



Answer the Question 5

☐ a

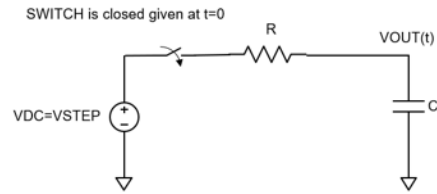
☐ c

☒ b

☐ d

Q6. Assume an ideal VDC: It can give any amount of current as required
 The system shown on the previous slide is subjected to an input step = VTSEP at t=0
 What is the initial slope (@t=0) of VOUT(t)?

- a) $\frac{V_{STEP}}{\tau}$, where $\tau = RC$
- b) $\frac{1}{\tau}$
- c) infinity
- d) 0



Answer the Question 6

- ☒ a

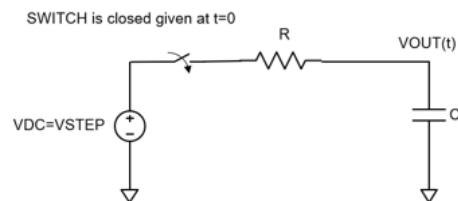
☐ c

☐ b

☐ d

Q7. Assume an ideal VDC: It can give any amount of current as required
 What is the current from VDC @t=0?

- a) $\frac{V_{STEP}}{R}$
- b) function of τ
- c) infinity
- d) 0



Answer the Question 7

- ☒ a

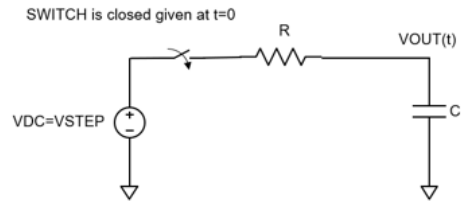
☐ c

☐ b

☐ d

Q8. Assume that the VDC is not ideal, and it can give only limited current, given by i_{limit} . Under what condition will $V_{OUT}(t)$ be i_{limit} limited?

- a) $V_{STEP} > i_{limit}/C$
- b) $\frac{V_{STEP}}{\tau} > i_{limit}/C$
- c) $\tau > i_{limit}/C$
- d) It will never be i_{limit} limited



Answer the Question 8

☐ a

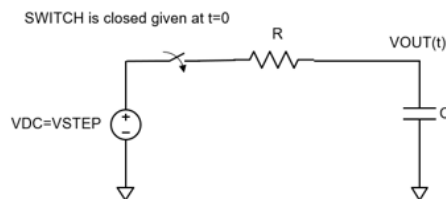
☐ c

☒ b

☐ d

Q9. Assume that the system initially gets limited, i.e. slope of $V_{OUT}(t) = i_{limit}/C$. At what value of remaining *error in voltage* = *final value* – *present value*, does it return to exponential behavior?

- a) $error = V_{STEP}$
- b) $Error = V_{STEP}/2$
- c) $Error = 0$
- d) $Error = (i_{limit}/C) * \tau$



Answer the Question 9

☐ a

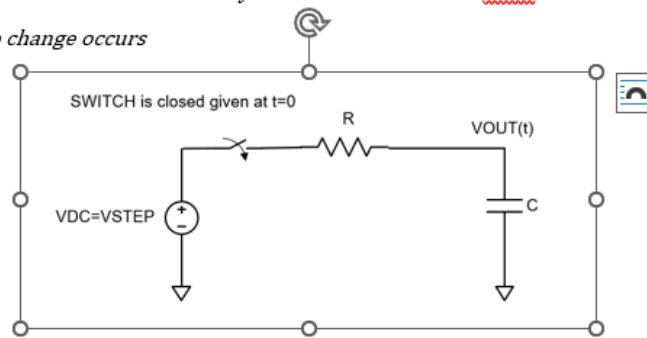
☐ c

☐ b

☒ d

Q10. If we reduce τ , while keeping i_{limit}/C constant, what happens?

- a) *Settling time always decreases*
- b) *i_{limit} limited component of settling time decreases*
- c) *System becomes more likely to become slow due to i_{limit}*
- d) *No change occurs*



Answer the Question 10

☐ a

☒ c

☐ b

☐ d

Done

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